Problem

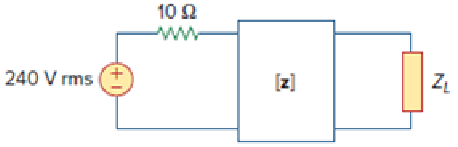
For the two-port circuit in Fig. 19.76,

$$[\mathbf{z}] = \begin{bmatrix} 40 & 60 \\ 80 & 120 \end{bmatrix} \Omega$$

(a) Find ***ZL*** for maximum power transfer to the load.

(b) Calculate the maximum power delivered to the load.

Figure 19.76



Step-by-step solution

Step 1 of 12

Refer to Figure 19.76 in the text book.

Consider the circuit diagram shown in Figure 1

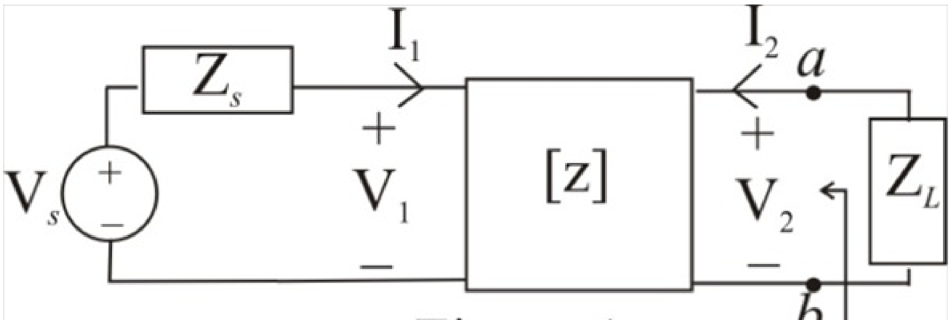


Figure 1

Step 2 of 12

Disconnect the load impedance and redraw the circuit.

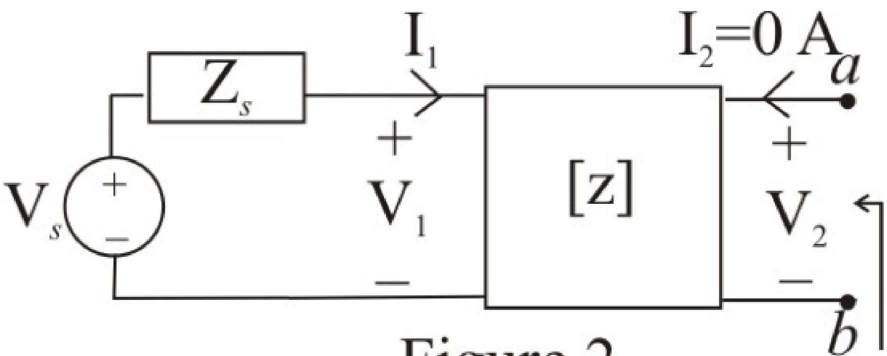


Figure 2

Step 3 of 12

Consider the **z**-parameter equation at output port.

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2$$

Substitute 0 A for $\mathbf{I}_2$.

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}(0)$$
$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1$$
$$\mathbf{I}_1 = \frac{\mathbf{V}_2}{\mathbf{z}_{21}}$$

Therefore, the expression fro the current $\mathbf{I}_1$ is $\frac{\mathbf{V}_2}{\mathbf{z}_{21}}$.

Step 4 of 12

Consider the **z**-parameter equation at input port.

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$

Substitute $\mathbf{V}_s - \mathbf{I}_1\mathbf{Z}_s$ for $\mathbf{V}_1$ and 0 A for $\mathbf{I}_2$.

$$\mathbf{V}_s - \mathbf{I}_1\mathbf{Z}_s = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}(0)$$
$$\mathbf{V}_s - \mathbf{I}_1\mathbf{Z}_s = \mathbf{z}_{11}\mathbf{I}_1$$
$$\mathbf{V}_s = (\mathbf{z}_{11} + \mathbf{Z}_s)\mathbf{I}_1$$

Substitute $\frac{\mathbf{V}_2}{\mathbf{z}_{21}}$ for $\mathbf{I}_1$.

$$\mathbf{V}_s = (\mathbf{z}_{11} + \mathbf{Z}_s)\left(\frac{\mathbf{V}_2}{\mathbf{z}_{21}}\right)$$
$$\mathbf{V}_2 = \mathbf{V}_s\left(\frac{\mathbf{z}_{21}}{\mathbf{z}_{11} + \mathbf{Z}_s}\right)$$

Here, the voltage $\mathbf{V}_s$ represents the Thevenin's voltage $\mathbf{V}_{Th}$.

$$\mathbf{V}_{Th} = \mathbf{V}_s\left(\frac{\mathbf{z}_{21}}{\mathbf{z}_{11} + \mathbf{Z}_s}\right) \quad \text{...... (1)}$$

Therefore, the value of the Thevenin's voltage $\mathbf{V}_{Th}$ across the terminals a-b is,

$\mathbf{V}_s\left(\frac{\mathbf{z}_{21}}{\mathbf{z}_{11} + \mathbf{Z}_s}\right)$.

Step 5 of 12

To find the Thevenin's impedance apply a voltage source at the output port and disconnect the source voltage as shown in Figure 3.

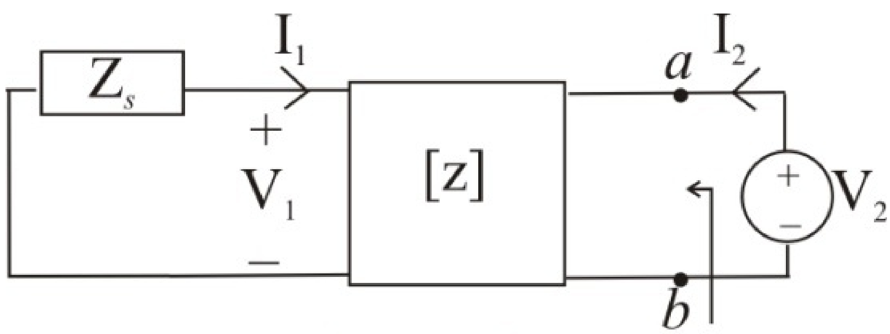


Figure 3

Step 6 of 12

Consider the **z**-parameter equation at input port.

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$

Substitute $-\mathbf{I}_1\mathbf{Z}_s$ for $\mathbf{V}_1$.

$$-\mathbf{I}_1\mathbf{Z}_s = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$
$$-\mathbf{I}_1\mathbf{Z}_s = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$
$$(\mathbf{z}_{11}+\mathbf{Z}_s)\mathbf{I}_1 = -\mathbf{z}_{12}\mathbf{I}_2$$
$$\mathbf{I}_1 = -\left(\frac{\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)\mathbf{I}_2$$

Therefore, the expression for the current $\mathbf{I}_1$ is $-\left(\frac{\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)\mathbf{I}_2$.

Step 7 of 12

Consider the **z**-parameter equation at output port.

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2$$

Substitute $-\left(\frac{\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)\mathbf{I}_2$ for $\mathbf{I}_1$.

$$\mathbf{V}_2 = \mathbf{z}_{21}\left(-\left(\frac{\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)\mathbf{I}_2\right) + \mathbf{z}_{22}\mathbf{I}_2$$
$$\mathbf{V}_2 = \left(\mathbf{z}_{22} - \frac{\mathbf{z}_{21}\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)\mathbf{I}_2$$
$$\frac{\mathbf{V}_2}{\mathbf{I}_2} = \mathbf{z}_{22} - \frac{\mathbf{z}_{21}\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}$$

The ratio $\frac{\mathbf{V}_2}{\mathbf{I}_2}$ represents the Thevenin's impedance $\mathbf{Z}_{Th}$.

$$\mathbf{Z}_{Th} = \mathbf{z}_{22} - \frac{\mathbf{z}_{21}\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s} \quad \ldots\ldots (2)$$

Therefore, the expression fro the current $\mathbf{I}_1$ is $\frac{\mathbf{V}_2}{\mathbf{z}_{21}}$.

(a)

Recall equation (2).

$$\mathbf{Z}_{Th} = \mathbf{z}_{22} - \frac{\mathbf{z}_{21}\mathbf{z}_{12}}{\mathbf{z}_{11}+\mathbf{Z}_s}$$

Substitute $10\ \Omega$ for $\mathbf{Z}_s$, $40\ \Omega$ for $\mathbf{z}_{11}$, $60\ \Omega$ for $\mathbf{z}_{12}$, $80\ \Omega$ for $\mathbf{z}_{21}$ and $120\ \Omega$ for $\mathbf{z}_{22}$.

$$\begin{aligned}\mathbf{Z}_{Th} &= 120 - \frac{(80)(60)}{40+10}\\ &= 120 - \frac{4800}{50}\\ &= 120 - 96\\ &= 24\ \Omega\end{aligned}$$

Therefore, the value of the Thevenin's impedance is, $24\ \Omega$.

To transfer the maximum power to the load, the value of the load impedance must be equal to the Thevenin's impedance. That is $\boxed{\mathbf{Z}_L = 24\ \Omega}$.

Step 8 of 12

(b)

Recall equation (1).

$$\mathbf{V}_{Th} = \mathbf{V}_s\left(\frac{\mathbf{z}_{21}}{\mathbf{z}_{11}+\mathbf{Z}_s}\right)$$

Substitute 240 for V_s, $10\ \Omega$ for $\mathbf{Z}_s$, $40\ \Omega$ for $\mathbf{z}_{11}$ and $80\ \Omega$ for $\mathbf{z}_{21}$.

$$\begin{aligned}\mathbf{V}_{Th} &= (240)\left(\frac{80}{10+40}\right)\\ &= (240)\left(\frac{80}{50}\right)\\ &= (24)(16)\\ &= 384\ \mathrm{Vrms}\end{aligned}$$

Therefore, the value of the Thevenin's voltage $\mathbf{V}_{Th}$ is $384\ \mathrm{Vrms}$.

Step 9 of 12

Draw the Thevenin's equivalent circuit diagram.

Step 10 of 12

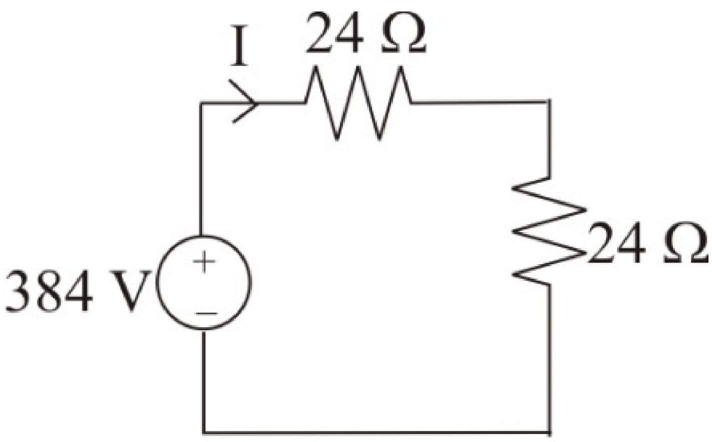


Figure 4

Step 11 of 12

Calculate the current passing in the circuit.

$$\begin{aligned}\mathbf{I} &= \frac{384}{24+24}\\ &= \frac{384}{48}\\ &= 8\ \mathrm{A\ rms}\end{aligned}$$

Therefore, the value of the current in the circuit is $8\ \mathrm{A\ rms}$.

Step 12 of 12

Calculate the value of the maximum power absorbed by the load.

$$P_{\max} = \mathbf{I}^2 \mathbf{Z}_L$$

Substitute $8\ \text{A rms}$ for $\mathbf{I}$ and $24\ \Omega$ for $\mathbf{Z}_L$.

$$\begin{aligned} P_{\max} &= (8)^2(24) \\ &= (64)(24) \\ &= 1.536\ \text{kW} \end{aligned}$$

Therefore, the value of the maximum power absorbed $P_{\max}$ by the load is $\boxed{1.536\ \text{kW}}$.